

Prajjawal Rajput

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SUMMARY

Full-stack developer building production web applications with TypeScript, React, and Node.js. Shipped a large-scale QR onboarding feature at Barco while building AI-powered workflows with LangChain, LangGraph, Agentic AI, and MCP.

EDUCATION

JSS Academy of Technical Education

B.Tech in Computer Science and Engineering — CGPA: 8.01

Noida, Uttar Pradesh

Oct 2023 – Present

Modern Era Public School

Class XII, PCM — 91%

Bijnor, Uttar Pradesh

Apr 2021 – May 2022

INTERNSHIP EXPERIENCE

Barco Electronic Systems Pvt. Ltd

Software Engineering Intern — MX Team (ClickShare & XMS Cloud)

Noida, Uttar Pradesh

Jun 2026 – Nov 2026

- Built and shipped a **Universal Web QR Code Scanner** in Angular/TypeScript from **POC to production** behind a **feature flag**, supporting onboarding of **1.53M+ ClickShare base units** (one unique QR per device) used across 500+ global companies.
- Reduced scan latency from **4+ seconds to under 1 second** by implementing a hybrid pipeline: native **BarcodeDetector** API (GPU-accelerated) with **zxing-wasm** fallback for iOS and Firefox, serving **253+ scans/day** in production.
- Improved scan reliability on difficult inputs using center-crop, contrast enhancement, nearest-neighbor upscaling, lens filtering, and adaptive zoom.
- Wrote **Mocha unit tests** for the Notification Manager, raising coverage from **64% to 99%** across all four metrics: statements, branches, functions, and lines.

PROJECTS

Pigeon | MERN, Socket.io, Zustand, Tailwind, Cloudinary, Arcjet, FastMCP | 🌐 GitHub | Live Jan 2026

- Built a full-stack real-time chat app with private 1-to-1 messaging, live online presence tracking, and JWT-based authentication secured via httpOnly cookies.
- Engineered a Socket.io WebSocket layer with per-socket JWT auth middleware for low-latency message delivery and real-time online/offline presence broadcasting across concurrent sessions.
- Integrated Arcjet rate limiting and bot protection, Cloudinary media uploads, and Mailtrap transactional emails for secure onboarding.
- Authored and deployed a Python FastMCP server exposing all 9 REST endpoints as LLM tools, enabling end users to connect from local Claude Desktop and perform authenticated chat actions.

AnyQuery AI | Next.js, FastAPI, LangChain, FAISS, Gemini | 🌐 GitHub | Live Jul 2026

- Built a full-stack RAG application that answers questions over YouTube transcripts and PDF documents using Next.js, FastAPI, LangChain, and Google Gemini, delivering grounded answers in **under 3 seconds** end-to-end.
- Implemented document ingestion with recursive chunking (**1000 chars, 200 overlap**), local **all-MiniLM-L6-v2** 384-dim embeddings, and FAISS top-**k=4** similarity search across **1000+ chunks** per document for grounded context retrieval.
- Developed a responsive chat interface with transcript language selection, PDF uploads, processing states, API error handling, and Markdown-rendered AI responses.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, TypeScript, SQL

Frontend: React.js, Next.js, Angular, Tailwind CSS, Zustand, Redux, TanStack Query

Backend: Node.js, Express.js, Socket.io, REST APIs

AI & Agent Frameworks: LangChain, LangGraph, MCP (Model Context Protocol), FastMCP, Agent Design Patterns

Databases: PostgreSQL, MongoDB, Drizzle ORM, Prisma ORM

Tools & DevOps: Git, GitHub, Sentry, Cloudinary, Clerk, Polar, WebAssembly

ACHIEVEMENTS

- Solved **400+ problems on LeetCode** and **600+ across platforms** spanning Arrays, Trees, Graphs, DP, and Backtracking